

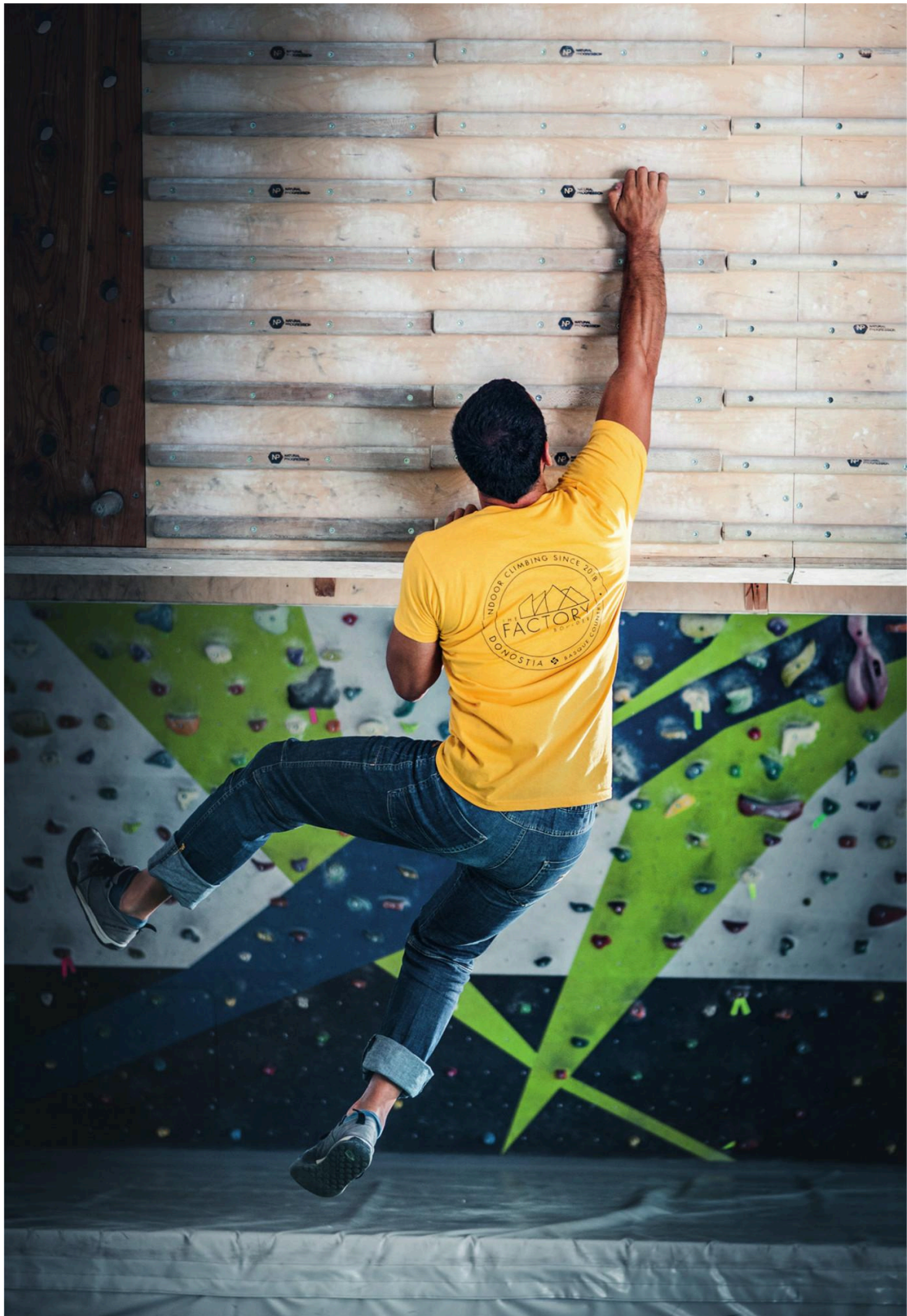
PART I

UNDERSTANDING TRAINING



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THE PROCESS OF TRAINING



DEFINITION

To begin, it's important to set a common understanding of what 'training' is. According to Bernal Ruiz (2006), training is a voluntary process that entails a transformation of physical and psychological functional systems. It occurs through the application of external stresses and in reaction to specific internal conditions in the body, and it's designed to improve performance in a particular situation.

Let's break this down into several key points.

The first thing to note is that training is a *process*; it is not a one-off session at the wall or the gym. It's also *voluntary*, meaning it's not going to happen while you're sitting on the sofa, scrolling through social media or talking about climbing at the pub. You have to get out there and train.

Then we get to the *transformation* of both *physical* and, especially in climbing, *psychological* systems. There's a clear and proven link between these two factors. Without sufficient physical training, no athlete will ever perform at their best. However, all the physical training in the world means nothing without the motivation, focus or resolution to win, take that next step, stick to the training or give that final push at the end of a race. And vice versa, regardless of how well prepared we are on a psychological level, without physical training, we'll never reach our full potential. What's more, with poor physical form, we start to doubt whether we can keep going or make that next move ... a vicious cycle that we'd do well to avoid.

The *external stresses* and *internal conditions* are two of the most important factors, and so we'll look at these in greater depth later on in the book. Basically, training must require genuine effort and the difficulty (technical, physical, psychological, and so on) of training exercises needs to vary according to the climber's level of training. For example, doing 10 pull-ups is not as hard for a F8a climber as it would be for a complete beginner whose previous hobbies amount to lying on the sofa and eating junk food.

The final thing to note is that our training should lead to a *higher level of performance*. If it doesn't, then something's gone wrong and we'll have wasted precious time and energy.

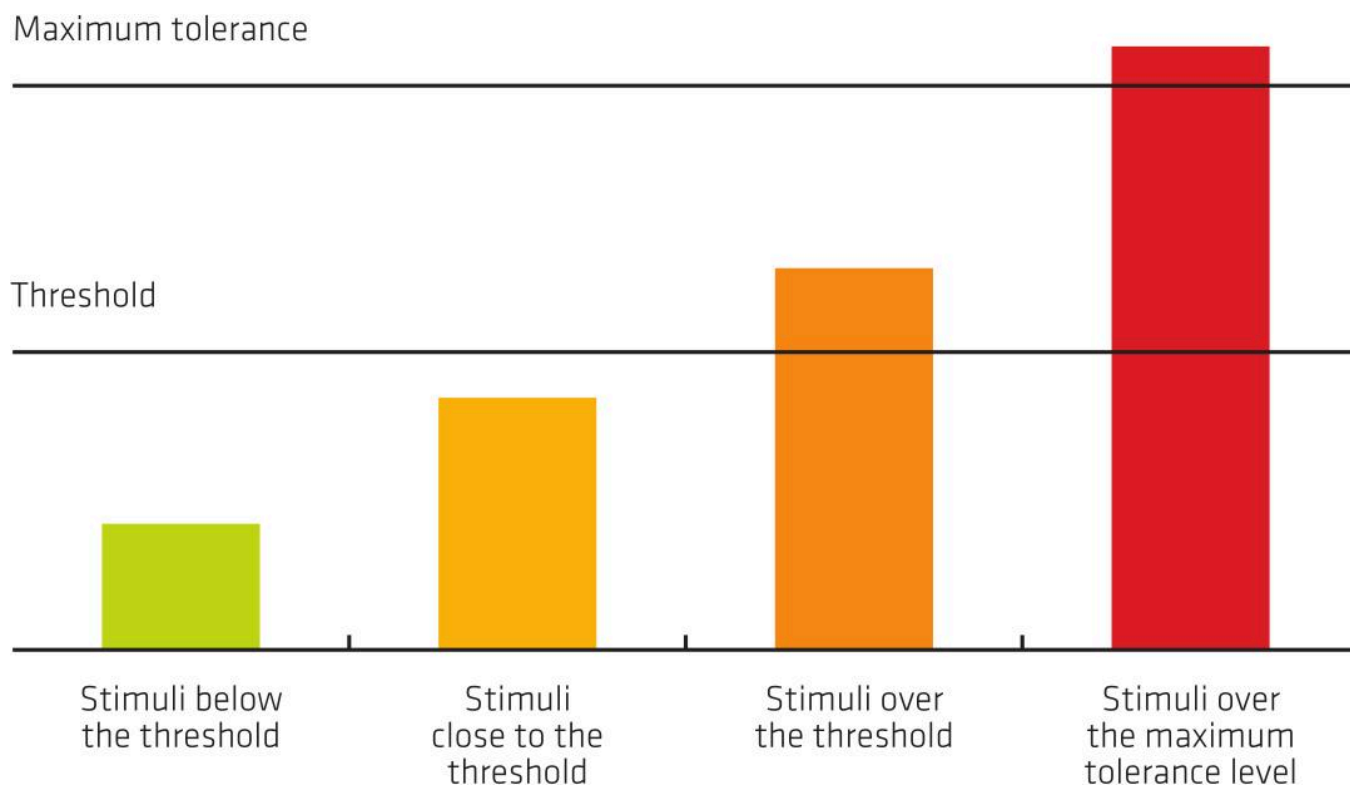
So, to get better at climbing, we just need to climb more and try harder routes? Unfortunately, it doesn't quite work like that, and it's the same for any sport: unless we're just starting out and don't climb very often, doing more won't make us any better and we won't climb any harder, no matter how hard we try.

Repeatedly doing any sport with no variation or evolution and, above all, no additional physical conditioning, leads to two possible scenarios: plateauing as we fail to master new techniques, or injury. If we never leave our comfort zone, where we climb and train comfortably, enjoy ourselves, feel proficient and capable, never try too hard or ever get frustrated, we'll never see any evolution or improvement because we're always doing the same thing. It would be like trying to train by brushing our teeth or drinking a glass of water. It's outside of this comfort zone where the magic happens: things evolve, changes occur ... However, without the necessary physical conditioning for the demands of our sport, we'll overload the most commonly used muscles, tendons, and so on. This can lead to excessive wear, agonist/antagonist imbalance and, most probably, to an injury that could stop us from climbing at all.

So, to climb better we need to train for climbing: we should dedicate our midweek sessions to training at the wall or the gym, and save climbing outdoors for the weekends (unlike many climbers who just tend to climb outdoors).

INTENSITY AND THRESHOLDS

How hard should I train? To answer this question, we must first understand the term 'threshold'. According to the *Oxford English Dictionary*, it is: 'a level, rate or amount at which something comes into effect'. Remember that functional adaptations are achieved by exposure to successively greater training stimuli. This is based on the Arndt-Schulz law, which basically states: 'stimuli that fall short of the threshold have no [training] effect'. This can be seen in the following graph:



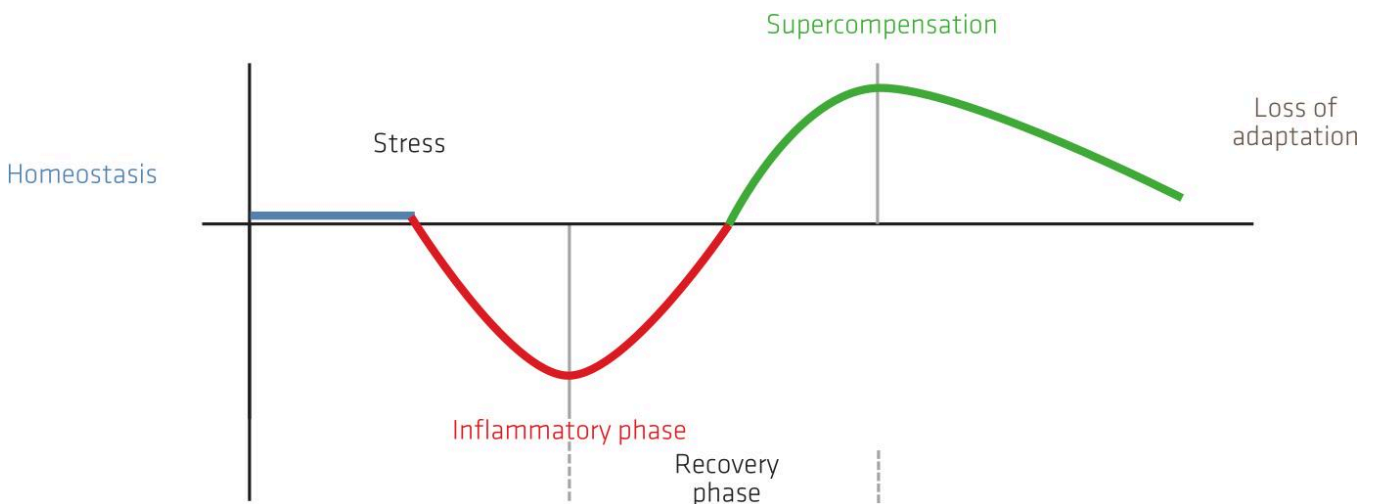
This graph demonstrates four possible scenarios:

- Stimuli below the threshold: no training effect. If our training doesn't require much effort, if it feels easy, then we're wasting our time.
- Stimuli close to the threshold: training effect if repeated, although some authors believe this only has a maintenance effect.
- Stimuli over the threshold: TRAINING EFFECT. These are the stimuli that we want. They take us out of our comfort zone and require just the right amount of effort. They make us dig deep and give that little bit extra.
- Stimuli over the maximum tolerance level: risk of overtraining syndrome and injury. If we need a few days to recover from a training session, we're probably over this level.

This should explain how hard to train: just hard enough, neither too little nor too much. As I said in the introduction: 'don't train more: train better'. Now let's look at how our bodies respond to stimuli that exceed the threshold to just the right degree, which is precisely what it means to 'train'.

HOMEOSTASIS, GAS AND SUPERCOMPENSATION

Homeostasis is when the body is in a state of equilibrium. In other words, when synthesis and degeneration are in balance (remember the body is in a constant state of regeneration: from our skin to the cells in our bones). If something, in this case training, disrupts homeostasis, the body will establish a new state of balance through regenerative or anabolic processes. This is known as general adaptation syndrome (GAS), a theory developed by Hans Selye.

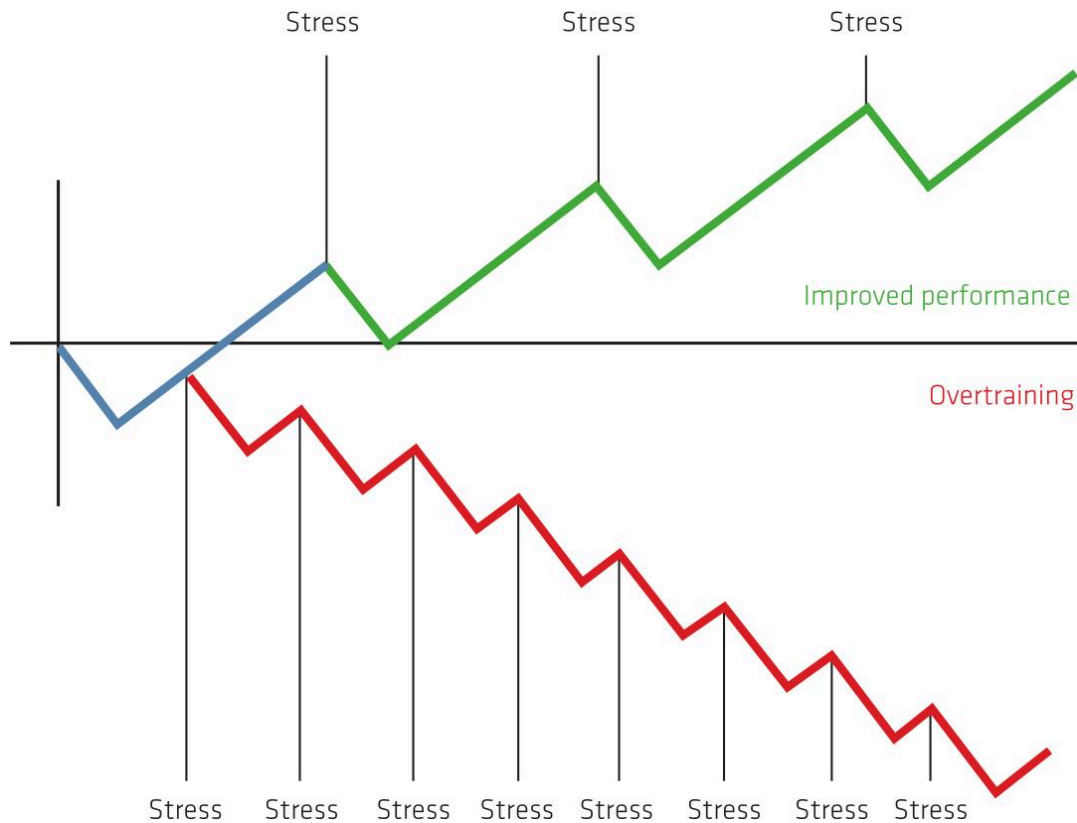


As seen in this graph, external stress disrupts the level of homeostasis in the body, leading to the inflammatory phase. During this phase, our body releases stress hormones such as adrenaline and steroid hormones such as corticosteroid (of which cortisol is the most well known). These hormones trigger physiological processes that allow us to withstand the *external stress* – the training – at the expense of 'damaging' our body, by creating what's known as *internal stress*.

When the external stress stops, the body tries to recover as quickly as possible. In other words, you're already recovering in the changing room after your session. To better withstand any similar external stress in the future, the body supercompensates to a heightened level of homeostasis, making you stronger than you were before the training session.

The bad news is that you'll lose the resulting adaptations if more stress isn't applied soon after the supercompensation phase. This is because the body adapts to the actual demands of our everyday lives. Logically, if something isn't needed and takes energy to maintain, the body will simply get rid of it to best adapt to our day-to-day needs. In even more bad news, the more extreme the adaptations (the harder you train), the faster they are lost because they're of least use. For example, the external stress that a sprinter would need to shave an extra hundredth of a second off a 100-metre sprint isn't really that useful for their body on a day-to-day basis. Training is like building a house: it can take years to build, but can be demolished in a matter of minutes.

That said, we have to be very careful about when we apply subsequent stress. If the body is still in the inflammatory phase, we risk entering a process called catabolism (degradation). If we don't realise this is happening, we could end up with overtraining syndrome and risk spending far more time at the doctor's than out at the crag, which is exactly what we don't want to happen. Luckily, it's difficult to fall into this trap unless you're training far too much and far too often.



THE PRINCIPLES OF TRAINING

The principles of training are a set of rules that help us to develop and manage our training. For me, the most important of these are:

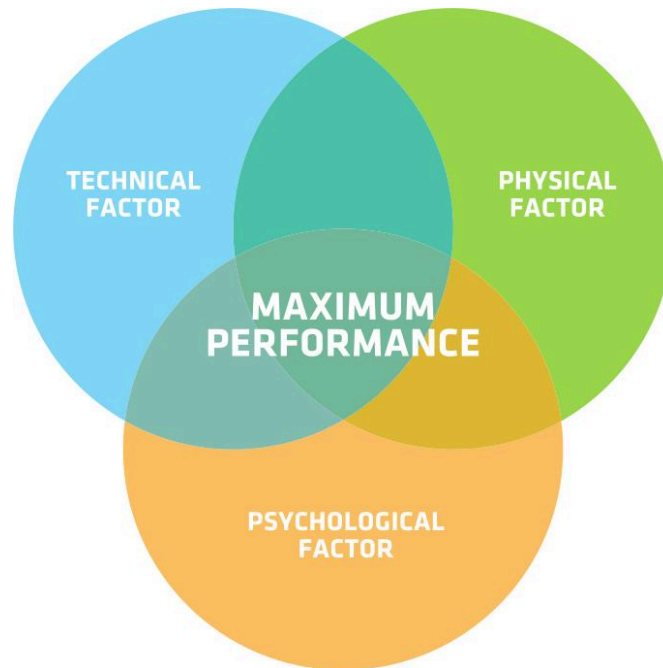
- *The principle of active and mindful participation:* perhaps the very reason behind this book. Understanding why we do things a certain way when training.
- *The principle of individualisation:* one that is often ignored. It means that any training should be adapted to the trainee's morpho-functional characteristics and level of ability. In other words, doing something just because it works for a friend is a big mistake.
- *The principle of specificity:* one of the most disheartening. It means that the gains from a certain exercise are not transferable to other situations. For example, training on a bike to improve aerobic capacity won't dramatically improve your aerobic capacity for long mountain walks. Similarly, the strength gained from doing pull-ups on a bar is only transferable to pulling down vertically on a bar-type hold.
- *The principle of continuity:* as the name suggests, only going to the gym once a month is simply a waste of membership fees.
- *The principle of variety:* training the same things in the same way is not going to work. You can't just repeat the same exercises for an entire season as the body will adapt and you'll see no improvement in performance. In the words of Albert Einstein: 'If you want different results [in our case, improved performance], do not do the same things.'

WHAT TO TRAIN

Now we know what training is, the logical thing to ask is: 'What should I train?'

In climbing, as in all sports, there are two types of training: visible training (physical factors) and invisible training (rest, nutrition, psychology, and so on). They are very closely linked and completely interdependent. Let's look at the specific factors that affect climbing performance:

- The technical factor (technique)
- The psychological factor (motivation, activation or arousal, fear, and so on)
- The physical factor (the physical condition of the climber)



The ability to maximise and synchronise each of these factors is key to achieving the best possible performance.

This book deals exclusively with the physical factor of climbing. If you want to explore the other two factors, there's lots of information out there about mental training and technique, although it's always best to seek advice from a qualified professional.

In the simple yet meaningful words of renowned American football coach Vince Lombardi: 'Fatigue makes cowards of us all.' We could say that in climbing, a sport where these factors are so deeply connected, you need good physical form not only for individual moves but to respond to the wider challenges of any given climb. For example, imagine you're totally spent on the ninth pitch of a route, graded say F7a, and there are still five pitches to go: one F7b+ and four at F6b or above. It's going to be a real fight to finish the climb. Imagine you're also in a pretty committed situation, where you can't just abandon the route ... things are really not looking good. This will of course affect your mental game. So, in this simple example, we've already touched on two of the three factors.

This book will analyse what to train to avoid these situations (at a higher or lower grade, on single or multi-pitch routes, alpine routes, boulder problems, and so on). Simply put, the physical factor is shaped by the following physical qualities: strength, endurance (cardiovascular), speed and flexibility. As you'll see throughout this book, strength and flexibility are the most important in climbing. Speed, in this context, is determined by strength and technique. Endurance, except when the approach to the crag is so strenuous that poor cardiopulmonary capacity will in fact affect climbing performance, is not a limiting factor *per se*. In any case, this will all be explored in greater depth.